

SQL Stored Procedures

A Stored Procedure is like a saved recipe for SQL — you write the steps once, give it a name, and just "call" that name whenever you want to run it again.

Basic Syntax:

```
CREATE PROCEDURE ProcedureName  
  
AS  
  
BEGIN  
    -- your SQL query goes here  
  
END
```

Stored Procedures with Parameters:

✓ 'IN' keyword

```
CREATE PROCEDURE SP_GetStudent  
    IN @ID INT                                (With 'IN' keyword)  
  
AS  
  
BEGIN  
    SELECT * FROM STUDENT WHERE STID = @ID;  
  
END
```

OR

```
CREATE PROCEDURE SP_GetStudent  
    @ID INT                                (Without 'IN' keyword)  
  
AS  
  
BEGIN  
    SELECT * FROM STUDENT WHERE STID = @ID;  
  
END
```

✓ 'OUT/OUTPUT' keyword

The OUT keyword in a stored procedure is used to return a value from the procedure back to the caller.

Basic Syntax:

```
CREATE PROCEDURE procedure_name
    @input_parameter datatype,
    @output_parameter datatype OUTPUT
AS
BEGIN
    -- statements
END;
```

Example:

```
CREATE PROCEDURE PR_count_employees(
    @city VARCHAR(20),
    @emp_count INT OUTPUT
)
BEGIN
    SELECT @emp_count=COUNT(*)
    FROM employees
    WHERE emp_city=@city
END
```

Execution:

```
DECLARE @count INT
```

```
EXECUTE PR_count_employees @City='Rajkot',@emp_count=@count OUTPUT
```

```
SELECT @count
```

The 4 Basic Operations:

1) Select

```
CREATE PROCEDURE SP_Insert  
  
AS  
  
BEGIN  
  
    SELECT * FROM STUDENT  
  
END
```

2) Insert

```
CREATE PROCEDURE SP_Insert  
  
    @Id INT, @Name VARCHAR(20)  
  
AS  
  
BEGIN  
  
    INSERT INTO STUDENT VALUES( @Id, @Name)  
  
END
```

3) Update

```
CREATE PROCEDURE SP_Update  
  
    @Id INT, @sem INT  
  
AS  
  
BEGIN  
  
    UPDATE STUDENT  
  
    SET SEM=@sem  
  
    WHERE STDID=@Id  
  
END
```

4) Delete

```
CREATE PROCEDURE SP_Delete  
    @Name VARCHAR(20)  
  
AS  
  
BEGIN  
  
    DELETE STUDENT  
  
    WHERE SNAME=@Name  
  
END
```

Execution of Stored Procedures:

Example-1:

If we have passed (id) as the parameter during creation of stored procedures (SP_Student) , then the execution will take place in any of the three ways listed below:

a) EXEC stored_procedure_name parameters

EXEC SP_Student 110

b) EXECUTE stored_procedure_name parameters

EXECUTE SP_Student 110

c) Stored_procedure_name parameters

SP_Student 110

Example-2:

If we have passed (id,name) as the parameters during creation of stored procedures (SP_Student) , then the execution will take place in any of the three ways listed below:

a) EXEC stored_procedure_name parameters

EXEC SP_Student 110, 'ABC'

b) EXECUTE stored_procedure_name parameters

EXECUTE SP_Student 110 , 'ABC'

c) Stored_procedure_name parameters

SP_Student 110, 'ABC'

DROP Command:

If we want to drop the given stored procedure, then:

Syntax:

DROP PROCEDURE stored_procedure_name

OR

DROP PROC stored_procedure_name

Example:

DROP PROCEDURE SP_Student